

Technology Stack

Date	26 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Technology Stack Details :

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3	Used to design a simple, responsive, and user-friendly interface for entering HDI parameters and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Flask provides a lightweight web framework for handling user requests, processing input data, and integrating the machine learning model efficiently.
3	Database / Data Storage	Pickle (.pkl Model File)	The trained machine learning model is stored as an HDI.pkl file, allowing fast loading and prediction without requiring a database.

4	Cloud / Hosting / Deployment	Render	Render provides free cloud hosting for deploying Flask applications with automatic deployment from GitHub.
5	Version Control & CI/CD	Git & GitHub	Git manages source code versions, while GitHub provides collaboration, backup, and automatic deployment integration with Render.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Pickle	Scikit-learn is used for model training and prediction, Pandas for data preprocessing, NumPy for numerical computations, and Pickle for saving/loading the trained model